# DHANUSH H A

+91-8618384324 | hadhanushsavitha@gmail.com | linkedin.com/in/dhanush-h-a | github.com/HADhanush

## Education

**B.M.S. College of Engineering** Bengaluru, Karnataka
*B.E, Electronics and Communication Engineering* *2022 – 2026*
CGPA: 8.50

**Sri Adichunchanagiri Science PU College** Arsikere, Karnataka
*Combination: PCMB* *2020 – 2022*
Percentage: 98

## Skills

**Languages:** Verilog, SystemVerilog, C/C++, Python
**Core Subjects:** Digital Design, CMOS VLSI Design, Verilog, System Verilog, Object Oriented Programming.
**Tools:** Xilinx Vivado, Intel Quartus Prime, ModelSim, Cadence Virtuoso, Synopsys Design Compiler (DC Shell), EDA Playground, VS Code

## Projects

**Convolutional Neural Networks (CNN's) Design Using Verilog** | *Python, Verilog*

- Dataset used : Kaggle ASL (American sign language) dataset
- Designed and implemented a Convolutional Neural Network architecture at RTL level using Verilog.
- Developed custom hardware modules for Kernal generation, Convolution and activation using fixed point arithmetic.
- Trained the ASL dataset using python tensorflow and extracted the weights and bias generated.
- Designed modular and parameterized RTL blocks enabling scalability and reuse across different CNN configurations.
- Verified functionality through simulation and testbenches, ensuring correctness of data flow and arithmetic operations.

**Synchronous FIFO Design Using Verilog :** | *Verilog*

- Developed a fully synthesizable synchronous FIFO to manage data flow between modules in the same clock domain.
- Designed control logic for read/write operations with pointer tracking and status flag generation (full, empty).
- Parameterized FIFO width and depth to support various system requirements.
- Verified functionality through testbenches and waveform analysis using Synopsys DC shell.

**Implementation and Performance Analysis of Sorting Algorithms on PYNQ-Z2 FPGA** | *Verilog HDL, FPGA*

- Implemented multiple sorting algorithms at RTL level using Verilog HDL, including Bubble Sort, Selection Sort, Quick Sort, Insertion Sort, and Radix Sort.
- Designed FSM based architectures to efficiently control sorting operations and manage data flow for array based inputs.
- Developed a top level hardware architecture to support batch wise sorting of incoming data streams.
- Analyzed hardware resource utilization such as LUTs and flip-flops, and optimized designs for improved performance.
- Integrated FPGA design with Python based control and testing using Jupyter Notebook on the PYNQ-Z2 platform.

## Additional Information

**Languages known:** Kannada, English, Hindi.

**Achievements:**

- Participated in an 8-hour hardware hackathon focused on building a line-following robot for automation.
- Participated in the e-Yantra Robotics Competition conducted by IIT Bombay and successfully completed Stage 1 tasks, gaining hands-on experience in Verilog.